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Inventor(s)

MANNHARDT; Joachim

DEVICE FOR EXAMINING MATERIAL SAMPLES BY MEANS OF ELECTROMAGNETIC RADIATION WITH SELECTABLE DETECTOR

Abstract

A device for examining material samples via electromagnetic radiation. The device comprises a lighting unit for generating the electromagnetic radiation with at least two radiation sources. The radiation of the radiation sources can be selectively directed onto the material sample. The device further comprises a detection unit having at least two detectors for capturing electromagnetic radiation emanating from the material sample. A deflection element is arranged in the detection unit, via which the electromagnetic radiation emanating from the material sample can be selectively deflected onto one of the detectors. This deflection element comprises a mirror, via which the radiation emanating from the material sample can be selectively deflected onto one of the detectors. The mirror is rotated about an axis extending substantially perpendicular to the optical axes of the detectors.

Inventors:	MANNHARDT; Joachim (Eschach, DE)
Applicant:	Nova Industrial Analytics GmbH (Eschach, DE)
Family ID:	88745709
Assignee:	Nova Industrial Analytics GmbH (Eschach, DE)
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Background/Summary

[0001] This nonprovisional application is a continuation of International Application No. PCT/EP2023/081138, which was filed on Nov. 8, 2023, and which claims priority to German Patent Application No. 10 2022 129 497.8, which was filed in Germany on Nov. 8, 2022, and which are both herein incorporated by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The invention relates to an apparatus for examining material samples via electromagnetic radiation.

Description of the Background Art

[0003] In many fields of industry pertaining to production and further processing, for example medical engineering, the food industry, etc., optical measuring methods are used to assess the state or quality of a product or an intermediate product. The term “optical measuring method” should be understood hereinafter to mean a measuring method using electromagnetic radiation, in particular electromagnetic radiation in a spectral range between infrared and ultraviolet. “Optical measurement” thus includes in particular a measurement in the far infrared (FIR), mid-infrared (MIR) and near infrared (NIR) spectral ranges, in the visible spectral range and in the UV range.

[0004] Such measurements are often associated with the desire to examine the material sample using electromagnetic radiation of different wavelengths. By way of example, for performing fluorescence measurements, biochemical samples can be labeled with two or more fluorophores, such that they emit light differently upon excitation with different light wavelengths. Biochemical properties can be measured in this way.

[0005] DE 699 29 086 T2 discloses a measuring device having a radiation source and a detector module, in which the radiation source transmits a combined beam made of UV and IR radiation, and the detector module comprises a plurality of individual detectors. By means of a rotary mirror arranged in the interior of the detector module, the radiation received by the detector module can be sequentially steered to the individual detectors without being split in the process.

SUMMARY OF THE INVENTION

[0006] The problem addressed by the present invention is that of developing the measuring apparatus known from the prior art, in such a way that an accurate assignment of the radiation measured in a predetermined spectral range to a spectral range of the excitation radiation is ensured. Furthermore, the apparatus should allow a quick and simple change between different spectral measurement regions.

[0007] An apparatus according to an example of the invention comprises an illumination device for generating electromagnetic radiation and a detection device having at least two detectors for capturing an electromagnetic radiation emanating from the material sample. The illumination

device comprises at least two radiation sources such that the radiation from one radiation source or the radiation from the other radiation source may be selectively directed at the material sample. The detection device comprises a deflection element by means of which the radiation emanating from the material sample may be selectively steered onto one of the detectors. According to the invention, this deflection element of the illumination device comprises a rotatable mirror by means of which the radiation may be selectively steered onto one of the detectors.

[0008] The design according to the invention of the apparatus is advantageous in that a plurality of radiation sources may be provided, the radiation of which may be used to illuminate the material sample. The apparatus according to the invention is particularly suitable for fluorescence analysis, in which the material sample is excited by polychromatic radiation or radiation in a limited spectral range, and the fluorescence radiation emitted by the material sample in a predetermined direction in the process is evaluated with the aid of the detection device.

[0009] In a first configuration of the invention, at least two of the radiation sources can have identical structures. This enables a redundant illumination, in particular in problematic surroundings, in which for example incandescent filaments of halogen lamps or anodes of mercury vapor lamps may crack or break off owing to mechanical loading, vibrations, etc.

[0010] Further, at least two of the radiation sources can have different structures, for example radiate in different spectral ranges, with different intensities, etc. This makes it possible to choose a radiation source that is optimally suitable for the respective measurement task. Furthermore, this allows a quick switchover between the radiation sources if radiation at different wavelengths should be successively steered onto the material sample for a given measurement task.

[0011] In particular, LED diodes transmitting in a predetermined spectral range may be used as radiation sources. In an alternative, use might be made of halogen lamps, for example, the radiation of which can be restricted to a predetermined spectral range by a filter.

[0012] The illumination device advantageously comprises a deflection element with the aid of which the radiation from one radiation source or the other may be selectively steered onto the material sample. This deflection element is designed such that it allows a quick switch-off or switch-on of the different radiation sources without complicated adjustment work being required within the scope of changing the radiation source.

[0013] The deflection element of the illumination device advantageously has beam-shaping properties. For example, the deflection element can be designed such that it focuses the radiation emitted by the respective radiation source onto the material sample; in particular, such a configuration is advantageous should the intention be to examine small objects. In an alternative, the deflection element may expand (defocus) the beam emitted by the radiation source, which may for example be advantageous in applications of reflection spectroscopy in the visible or infrared spectral range, when the intention is to illuminate the material sample over a large area.

[0014] The deflection element of the illumination device may comprise a mirror in particular, by means of which the one or the other radiation source may be selectively steered onto the material sample. For example, use may be made of a concave mirror, in particular a parabolic mirror, which has a collimation effect that is advantageous for spectroscopic measurements.

[0015] Expediently, the deflection element of the illumination device is rotatably mounted, to be precise in such a way that the axis of rotation is aligned parallel to the propagation direction of the deflected radiation. In this case, the two or more radiation sources may be arranged on a circular arc around the deflection element, and rotations of the deflection element allow the radiation from the one or the other radiation source to be selectively steered onto the material sample without further adjustments being required.

[0016] The deflection element of the illumination device may have an opening for passing through electromagnetic radiation. This is expedient, in particular, should the spectrum of the intensity of the radiation reflected off the material sample in the direction of incidence be measured, for example within the scope of determining the intensity or checking the functionality in transmission

measurements. Furthermore, it is advantageous to provide an optical waveguide, in particular a light-guiding rod, which guides the radiation reflected off the material sample in the direction of incidence through the deflection element of the illumination device with low losses, in the region of this passage opening.

[0017] To allow fluorescence measurements in different spectral ranges in quick temporal succession, it is advantageous to synchronize the switch-on and switch-off of the various radiation sources of the illumination device with the switch-on and switch-off of the different detectors of the detection device. In this way, a plurality of measurements in different spectral ranges may be performed on the material sample in quick succession.

[0018] If, as described above, the illumination device is provided with a deflection element by means of which radiation from a specific radiation source may be selectively steered onto the material sample, then it is advantageous to synchronize the movements of this deflection element with the movements of the deflection element of the detection device such that the radiation source active at a given time corresponds to the associated detector.

[0019] In a manner analogous to the configuration of the deflection element of the illumination device, the deflection element of the detection device may also have beam-shaping properties by way of for example focusing the radiation emanating from the material sample onto the detectors.

[0020] Further scope of applicability of the present invention will become apparent from the detailed description given hereinafter. However, it should be understood that the detailed description and specific examples, while indicating preferred embodiments of the invention, are given by way of illustration only, since various changes, combinations, and modifications within the spirit and scope of the invention will become apparent to those skilled in the art from this detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The present invention will become more fully understood from the detailed description given hereinbelow and the accompanying drawings which are given by way of illustration only, and thus, are not limitive of the present invention, and wherein:

[0022] FIG. 1 shows a schematic sectional view of an apparatus for examining material samples, having a plurality of light sources and a plurality of detectors;

[0023] FIG. 2 shows a schematic sectional view of a further apparatus for examining material samples, having a plurality of light sources and a plurality of detectors; and

[0024] FIG. 3 shows a schematic sectional view of an alternative apparatus for examining material samples, having a plurality of light sources and a plurality of detectors.

DETAILED DESCRIPTION

[0025] FIG. 1 shows a schematic sectional illustration of an apparatus **10** for examining a material sample **40** by means of fluorescence spectroscopy. The apparatus **10** comprises an illumination device **20** for generating the electromagnetic radiation and a detection device **70** for capturing the electromagnetic radiation emanating from the material sample **40**. The illumination device **20** is tilted vis-à-vis the material sample **40** at an angle of incidence **80**, for example 45° , and so the radiation **28** emanating from the illumination device **20** is incident on the material sample **40** at the angle of incidence **80**. The detection device **70** is also inclined vis-à-vis the material sample **40** such that the intensity of the radiation **78** emitted by the material sample **40** is measured at an angle **80'**.

[0026] The illumination device **20** comprises two radiation sources **21**, **21'** in the form of halogen lamps that are provided with different wavelength-selective filters **32**, **32'**, whereby the emanating radiation **22**, **22'** has different spectral properties. The two radiation sources **21**, **21'** are arranged

diametrically opposite one another such that their respective optical axes are aligned flush with one another. A deflection element **23** arranged in the beam path between the two radiation sources **22**, **22'** serves to selectively steer the beams **22** of the first radiation source **21** or the beams **22'** of the second radiation sources **21'** onto the material sample **40**. The deflection element **23** comprises a mirror **24**, a concave mirror **24'** in this example, which may be rotated about an axis of rotation **26** that is perpendicular to the propagation direction of the beams **22**, **22'**.

[0027] In FIG. **1**, the mirror **24** is in a position in which the radiation **22** from the first radiation source **21** is deflected with the aid of the mirror **24** and focused onto the material sample **40**, while the radiation **22'** from the second radiation source **21'** is blocked by the deflection element **23**. With the aid of the mirror **24**, the radiation **21** from the first radiation source **21** is reflected into a propagation direction **27** that extends parallel—and collinearly in the present example—to the axis of rotation **26** of the deflection element **23**. With the aid of a drive unit **35**, the mirror **24** may be rotated through 180° from the position shown in FIG. **1** such that radiation **22'** from the second radiation source **21'** reaches the material sample **40**, while the radiation **22** from the first radiation source **21** is blocked by the deflection element **23**. In an alternative to the halogen lamps shown in FIG. **1**, e.g. LEDs with a narrow spectral emission range may be used as radiation sources **21**, **21'**. In addition to the two halogen lamps, further radiation sources may be provided, which are preferably arranged on a great circle around the axis of rotation **26** of the deflection element **23**.

[0028] The detection device **70** comprises two detectors **71**, **71'** that are arranged diametrically opposite one another in the example of FIG. **1** such that their respective optical axes are aligned flush with one another. The two detectors **71**, **71'** serve to measure the intensity of a radiation **78**, which is incident on the detection device **70** in a direction of incidence **78**, in different spectral ranges. To this end, use can be made of detectors **71**, **71'** that inherently measure in different spectral ranges. In an alternative, use can be made of detectors **71**, **71'** with an identical construction and a broad spectral sensitivity range that have been provided with different wavelength-selective filters **79**, **79'** such that the radiation **78** is filtered in wavelength-specific fashion before it reaches the detectors **71**, **71'**.

[0029] Arranged between the two detectors **71**, **71'** is a deflection element **73**, with the aid of which a radiation **78** incident on the detection device **70** may be selectively steered onto one of the two detectors **71**, **71'**. The deflection element **73** comprises a mirror **74**, a concave mirror **74'** in this example, which may be rotated about an axis of rotation **76** that is parallel to the incident radiation **78**. A drive unit **75** is provided for rotating and positioning the mirror **74**. In FIG. **1**, the mirror **74** is in a position in which the radiation **78** emitted by the material sample **40** is focused via the mirror **74** onto the detector **71** (beam **72**), while the second detector **71'** is shielded from the radiation **78** by the deflection element **73**. A rotation of the mirror **74** through 180° steers the radiation **78** onto the second detector **71'** (beam **72'**), while the first detector **71** is now shielded. In this way—by rotating the mirror **74**—it is possible to use the detectors **71**, **71'** to measure the intensity of the radiation **78** in different wavelength ranges. In addition to the detectors **71**, **71'** shown in FIG. **1**, even more detectors may be present, said further detectors being arranged in a great circle in a plane perpendicular to the axis of rotation **76** of the mirror **74**. A drive unit **75** is provided for the positionally accurate rotation of the mirror.

[0030] The illumination device **20** and the detection device **70** are sealed from the surroundings by way of observation windows **31**, for example sapphire windows, in order to suppress the ingress of dust and other contamination into the interior.

[0031] Advantageously, the wavelength-selective filters **32**, **32'** of the illumination device **20** and the wavelength-selective filters **79**, **79'** of the detection device are matched to one another in pairs so that the material sample **40** can be examined in two different spectral ranges without apparatus-based outlay, simply by rotating the mirrors **24**, **74**. Furthermore, a synchronized rotation of the two mirrors **24**, **74** allows switching back and forth between the two different spectral measurement regions in quick temporal succession. In this case, the mirrors **24**, **74** can be rotated continuously or

incrementally. Thus, a continuous rotation of the mirrors **24**, **74** allows radiation from the one or the other radiation source **21**, **21'** to be steered onto the material sample **40** at regular time intervals and be detected wavelength-selectively in synchronized fashion by means of the detectors **71**, **71'**. In an alternative, the mirrors **24**, **74** may be rotated in positioning fashion.

[0032] FIG. **2** shows a further apparatus **10'** for examining a material sample **40** with the aid of the illumination device **20** and the detection device **70**, both of which are arranged flush and perpendicular to the material sample **40** in this example. The deflection element **23** of the illumination device **20** is provided with an opening **25** for receiving a light-guiding rod **36**, by means of which a radiation reflected off the material sample **40** in the direction of incidence **27** can be transmitted through the deflection element **23** in the direction of the detection device **70**. Using the illumination device **20**, the material sample **40** is selectively or temporally alternately illuminated with radiation from the radiation source **21** or the radiation source **21'**. The radiation **78** reflected off the material sample **40** in the direction of incidence **27** is guided through the deflection element **23** of the illumination device **20'** by means of a light-guiding rod **36** and is incident on the detection device **70**, in which the radiation **78** is selectively or temporally alternately steered onto the detector **71** or the detector **71'**. Should measurements in different spectral ranges be performed in quick alternation, the movements of the deflection elements **23**, **73** in the illumination device **20'** and in the detection device **70** are advantageously synchronized in such a way that for a given radiation **22**, **22'** in the detection device, the detector **71**, **71'** in each case corresponding to the spectral range of this radiation **22**, **22'** is enabled.

[0033] FIG. **3** shows an example of an apparatus **100** according to the invention, in which the detection device **70** substantially corresponds to the detection device shown in FIGS. **1** and **2**. The illumination device **120** comprises a plurality of LEDs **121**, **121'** with different spectral properties, which are arranged on a great circle in a ring around the material sample **40** such that the radiation **122**, **122'** of said LEDs is aligned with the material sample **40**. The material sample **40** is impinged upon with radiation at different wavelengths by an alternating switch-on and switch-off of these LEDs **121**, **121'**. The detection device **70** comprises detectors **71**, **71'** whose spectral properties are matched to the spectra of the LEDs **121**, **121'**.

[0034] The alternating switching of the LEDs **121**, **121'** can be achieved with the aid of a rotatable disk **130** in particular. The disk **130** has a central cutout **131** for the radiation **78** emanating from the material sample **40**. Furthermore, a lateral cutout **132** is provided on the disk **130** and it allows the radiation from in each case one of the LEDs (the LED **121** in the present case) to be incident on the material sample **40**. The angular position of the disk **130** is synchronized with the angular position of the mirror **74** in the interior of the detection device **70**. In this way, the material sample **40** may be successively irradiated by different LEDs, and via the mirror **74** a measurement of the radiation emitted by the material sample **40** may be performed at the same time by the detector **71** assigned to this LED. In this case, the disk **130** may be electrically or mechanically linked to the drive unit **75** of the mirror **74**.

[0035] The invention being thus described, it will be obvious that the same may be varied in many ways. Such variations are not to be regarded as a departure from the spirit and scope of the invention, and all such modifications as would be obvious to one skilled in the art are to be included within the scope of the following claims.

Claims

1. An apparatus for examining a material sample via electromagnetic radiation, the apparatus comprising: an illumination device to generate the electromagnetic radiation, the illumination device comprising at least two radiation sources, radiation of which being adapted to be selectively directed at the material sample; and a detection device having at least two detectors to capture an electromagnetic radiation emanating from the material sample, the detection device comprising a

- deflection element via which the radiation emanating from the material sample is adapted to be selectively steered onto one of the detectors, wherein the deflection element of the detection device comprises a mirror via which the radiation emanating from the material sample is adapted to be selectively steered onto one of the detectors, and wherein the mirror rotatable about an axis that extends substantially perpendicular to optical axes of the detectors.
2. The apparatus as claimed in claim 1, wherein the at least two of the radiation sources have substantially identical structures.
 3. The apparatus as claimed in claim 1, wherein the at least two of the radiation sources have different structures.
 4. The apparatus as claimed in claim 1, wherein the at least two radiation sources are LED diodes.
 5. The apparatus as claimed in claim 1, wherein the illumination device comprises a rotatable disk which is provided with cutouts and via which radiation from one of the radiation sources is adapted to be selectively steered onto the material sample.
 6. The apparatus as claimed in claim 5, wherein the respective angular position of the disk of the illumination device is adapted to be synchronized with a respective position of the deflection element of the detection unit.
 7. The apparatus as claimed in claim 1, wherein the illumination device comprises a deflection element via which radiation from one of the radiation sources is adapted to be selectively steered onto the material sample.
 8. The apparatus as claimed in claim 7, wherein the deflection element of the illumination device has beam-shaping properties.
 9. The apparatus as claimed in claim 7, wherein the deflection element of the illumination device comprises a mirror via which radiation from one of the radiation sources is selectively guided onto the material sample.
 10. The apparatus as claimed in claim 7, wherein the deflection element is rotatable about an axis that is substantially parallel to a propagation direction of the deflected radiation.
 11. The apparatus as claimed in claim 7, wherein the deflection element has an opening for the passage of electromagnetic radiation.
 12. The apparatus as claimed in claim 11, wherein a light-guiding rod is arranged in the region of the opening.
 13. The apparatus as claimed in claim 7, wherein respective settings of the deflection element of the illumination device and of the deflection element of the detection unit are synchronizable.
 14. The apparatus as claimed in claim 1, wherein the deflection element of the detection device has beam-shaping properties.
 15. The apparatus as claimed in claim 1, wherein the mirror is a concave mirror.
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